

Lesson 9 Teacher's Checklist / Prep Guide

Complete before class

VIDEO GAME CREATION

This Week: The Payoff of Deferring UI

Since Lesson 6, every engine's win/lose messages have deliberately used the simplest possible one-off text tool, with a promise that "the real UI system comes later, once there's a score and lives to actually display." This week is that payoff. Some engines (Godot, GameMaker, GDevelop, TIC-80, Scratch) already had the right tool the whole time - their Lesson 6/8 text mechanism just needs to run continuously instead of once. Two engines (Unity, Unreal) are genuinely introducing a new system this week: Unity's UI Text + Canvas, and Unreal's UMG Widget Blueprint. 001 Game Creator also introduces something new: an Interface with bound Fields, replacing one-off Message Boxes for this purpose.

- ☐ Read through each engine's Handout once before class - Unity, Unreal, and 001 Game Creator involve genuinely new UI concepts this week, more setup than a typical lesson.

001 Game Creator: Verified Mechanism

001 Game Creator's HUD approach (an Interface containing Text/Graphic Fields, bound to a Variable via "Retrieve Text/Value" with "Update Real-time" ticked) was verified directly against the official wiki (001gamecreator.com/wiki/field/) before this lesson was written. The most important checkbox to not miss: Update Real-time. Without it, the Interface only refreshes when reopened, which looks like a bug but is really just a missed setting.

- ☐ If you have Windows access before class, walk through creating one Interface with one bound Field yourself - this is new territory even for an instructor comfortable with 001 Game Creator's trigger-based patterns.

Two Update Patterns Worth Naming

This lesson quietly introduces two different, both-correct ways to keep a display in sync with data: PUSH (call an update function every time the data changes - used by Godot, GDevelop, 001 Game Creator's real-time binding, Unreal) and POLL (just re-check the data every single frame, no matter what - used naturally by GameMaker's Draw GUI event, TIC-80's print() calls, and this week's Unity HUD script). Neither is "more correct" - it's worth drawing this contrast out loud if a strong student asks why the approaches differ across engines.

Self-Contained Lesson Assets

No new assets this week - the HUD is built from each engine's own text/UI tools.

Continuity From Lesson 8

- ☐ Remind students to open their own [YourName]Game project from Lesson 8, not create a new one.
- ☐ If a student's Lesson 8 project is lost or broken, have a pre-built "Lesson 8 complete" starter available per engine (from the Lesson 8 Teacher's Keys) so they aren't blocked.

Engine Verification

- ☐ Confirm the demo laptop's Lesson 6 and 8 exercises still work for all 8 engines.
- ☐ Unity: confirm GameObject > UI > Text (Legacy) is available and creates a Canvas automatically - some Unity versions nudge toward TextMeshPro first; Legacy Text is intentionally used here to avoid an extra package-import step.
- ☐ Unreal: confirm Widget Blueprint creation and "Create Widget" / "Add to Viewport" nodes work smoothly on the demo laptop - this is the most setup-heavy engine this week.
- ☐ 001 Game Creator (Windows demo laptop): walk through Interface creation once, and specifically confirm the Update Real-time checkbox's effect is visible (toggle it off briefly to show students what "not real-time" looks like, if time allows).
- ☐ Godot / GameMaker / GDevelop / TIC-80 / Scratch: confirm score increments by exactly 10 per kill and the on-screen display updates immediately, every time, with no stale numbers.

Materials

- ☐ Print or queue digital copies of all 8 Student Handouts (or confirm students can access their engine's version electronically).
- ☐ Have the 8 Teacher's Keys on hand (instructor reference only - do not distribute) in case a student needs a worked example beyond Level 1.
- ☐ Bring: demo laptop, projector cable/adapter, handout copies.

Known Gotchas This Week

- 001 Game Creator: forgetting Update Real-time is the single most likely mistake - the Interface will look broken (frozen numbers) even though the underlying Variable is updating correctly.
- Unity / Unreal: students may accidentally create multiple Canvas or Widget instances by repeating setup steps - if a HUD element appears duplicated or doubled, check for this first.
- Every engine: a score readout stuck at 0 usually means the increment was added to the wrong code path (or forgotten entirely), not a display bug - have students confirm the underlying variable is changing (via a temporary debug print/log if needed) before troubleshooting the display itself.